Lebanese University
Faculty of Sciences
Section V

Second Session
Duration: 50 mins

Applied Mathematics and Statistics for Computational Sciences

---

Problem 1: Principal Component Analysis

A dataset contains $n = 150$ observations described by three centered and standardized quantitative variables:

$$X_1 = \text{execution time}, \qquad X_2 = \text{memory consumption}, \qquad X_3 = \text{accuracy score}.$$

A Principal Component Analysis is performed on the dataset.

The eigenvalues of the correlation matrix are:

Component	PC1	PC2	PC3
Eigenvalue	1.80	0.90	0.30

The first two principal components are given by:

$$\text{PC1} = 0.65X_1 + 0.68X_2 - 0.34X_3 \quad ; \quad \text{PC2} = 0.25X_1 + 0.22X_2 + 0.94X_3.$$

1) Verify that the sum of the eigenvalues is equal to the total variance of the standardized variables.

2) Compute the percentage of total variance explained by each principal component.

3) Compute the cumulative percentage of variance explained by the first two principal components.

Based on this calculation, how many principal components would you retain? Justify your answer.

4) Interpret PC1 using the signs and magnitudes of its coefficients.

What type of observations are expected to have a high positive value on PC1?

5) An observation has the following standardized values:

$$X_1 = 1, \qquad X_2 = 0.5, \qquad X_3 = -1.$$

Compute its coordinates on PC1 and PC2.

6) After projection onto the first two principal components:

(a) What is the dimension of the transformed dataset?

(b) What percentage of total information is lost?

Problem 2: Optimization and Gradient Descent

Consider the function

$$f(x) = (x - 4)^2 + 2.$$

1) Compute the derivative $f'(x)$.

2) Determine analytically the value $x^\star$ that minimizes f, and compute the corresponding minimum value $f(x^\star)$. **3)** The gradient descent update rule is

$$x^{(k+1)} = x^{(k)} - \eta f'\left(x^{(k)}\right).$$

Starting from

$$x^{(0)} = 0$$

and using the learning rate

$$\eta = 0.25,$$

compute

$$x^{(1)}, \qquad x^{(2)}, \qquad x^{(3)}.$$

For each iteration, also compute the corresponding value of the objective function.

Complete the following table:

Iteration k	$x^{(k)}$	$f\left(x^{(k)}\right)$
0	0	
1		
2		
3		

4) Consider the following incomplete Python code:

```
def f(x):
    return (x - 4)**2 + 2

def gradient(x):
    return ______________________

x = 0.0
learning_rate = 0.25
tolerance = 1e-4
max_iterations = 100

for iteration in range(max_iterations):
```

```
    old_x = x

    x = ______________________________

    if abs(x - old_x) < tolerance:
        break

print("Minimum point:", x)
print("Minimum value:", f(x))
```

(a) Complete the two missing lines.

(b) Explain the role of `learning_rate`.

5) The condition

```
abs(x - old_x) < tolerance
```

means that the algorithm stops when:

(A) the value of the function becomes zero;

(B) two successive iterates become sufficiently close;

(C) the current value of x becomes zero;

(D) the gradient becomes positive.

6) Which of the following statements is correct?

(A) Gradient descent always converges in one iteration.

(B) A very large learning rate always improves convergence.

(C) Gradient descent updates the current point in the direction opposite to the gradient.

(D) The gradient is always equal to the value of the objective function.